B4 aligned coordinates: z = 0.5 mm

Offline exact frequency-domain alignment; original frequency bins, no smoothing or fractional-delay implementation.

B4 validation LRM — B4 fitted model

Rx → Rx | Ry → Rx | Z → Rx | N → Rx

Rx → Ry | Ry → Ry | Z → Ry | N → Ry

Rx → Z | Ry → Z | Z → Z | N → Z

Rx → N | Ry → N | Z → N | N → N

Magnitude (dB, recorded normalized coordinates)

Frequency (Hz)